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1. Implementation Process

The implementation process started after the main idea of the Tank Battle Game was selected. I first created the basic game window using Java Swing. After that, I added the player tank and keyboard controls so the tank could move in four directions. Then I added bullets so the player could shoot.

After the player movement and shooting worked, I added enemy tanks. The enemy tanks move automatically by using simple random direction changes. I also added random enemy shooting to make the game more challenging. Then I added walls, collision detection, score, health, winning, losing, and restart logic.

The project follows object-oriented programming by separating the main objects into classes. Tank is the parent class, while PlayerTank and EnemyTank inherit from it. Bullet and Wall are separate classes because they have their own responsibilities.

1.1 System Requirements

- Java Development Kit installed.
- Any computer that can run Java.
- Keyboard for controlling the tank.
- IntelliJ IDEA, Eclipse, VS Code, or command line for running the program.

1.2 User Install Guide

- Download or copy the file TankBattleGame.java.
- Open the file in IntelliJ IDEA or another Java IDE.
- Make sure the file name is exactly TankBattleGame.java.
- Run the main method in the TankBattleGame class.
- The game window will open and the player can start playing.

2. Changes and Improvements

At first, the project only had a basic tank moving on the screen. Later, more features were added to make the game complete. Enemy tanks, bullets, walls, health, score, and restart were added during implementation.

2.1 Enemy Improvements

Enemy tanks were added so the player has a real challenge. Enemies move randomly and shoot bullets at random times. This makes the game different each time it is played.

2.2 Bullet Improvements

Both the player and enemy tanks can shoot bullets. Bullets move in the direction of the tank and disappear when they go outside the screen or hit a wall.

2.3 Wall Improvement

Walls were added as obstacles. Tanks cannot pass through them, and bullets are stopped by them. This makes the map more interesting and adds strategy to movement.

2.4 Score and Health Improvement

The game displays health, score, and enemies left at the top of the screen. The score increases when an enemy tank is destroyed. The game ends when the player health reaches zero.

3. User Manual

3.1 How to Start

The user starts the game by running the TankBattleGame.java file. The game window opens automatically.

3.2 How to Play

- Press W to move up.
- Press S to move down.
- Press A to move left.
- Press D to move right.
- Press Space to shoot.
- Press R to restart the game.

3.3 How to Win

The player wins by destroying all enemy tanks. When all enemies are destroyed, the game shows the You Win message.

3.4 How to Lose

The player loses if enemy bullets reduce the player health to zero. When this happens, the game shows the Game Over message.

4. What is Missing?

- The game does not have sound effects yet.
- The game does not have different levels yet.
- The game does not save high scores yet.
- The game uses simple shapes instead of image graphics.
- The game does not have multiplayer mode yet.

5. Conclusion

The Tank Battle Game was successfully implemented as a Java object-oriented game. It includes player movement, enemy tanks, shooting, walls, health, score, win condition, game over condition, and restart. The project is simple enough to understand but complete enough to show important software engineering and programming concepts.